

ENGINEERING

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EXPLORATION

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WHAT IS ENGINEERING?

Engineering is the practical use of science and math to design, build, and improve things that solve real-world problems. It covers everything from bridges and buildings to machines, electronics, software, and manufacturing processes



DIFFERENCE BETWEEN SCIENCE AND ENGINEERING

- **SCIENCE**

Science is the systematic study of the natural world through observation, experimentation, and logical reasoning to understand how things work.



- **ENGINEERING**

Engineering is the application of scientific and mathematical principles to design, build, test, and improve machines, structures, systems, and technologies that solve real-world problems



DIFFERENCE BETWEEN SCIENTIST AND ENGINEER.

- **SCIENTIST**

A scientist is a person who studies the natural world through observation, experiments, and research to discover new knowledge and explain how things work.



- **ENGINEER**

An engineer is a professional who uses scientific, mathematical, and technical knowledge to design, build, test, and improve machines, structures, systems, and technologies to solve practical problems



NEEDS AND WANTS OF THE COMPUTER SCIENCE ENGINEER IN TODAY'S WORLD

- **NEEDS (ESSENTIAL REQUIREMENTS)**

1. Strong programming skills (C, C++, Java, Python, etc.).
2. Knowledge of data structures and algorithms.
3. Understanding of databases, operating systems, and computer networks.
4. Problem-solving and analytical thinking.
5. Communication and teamwork skills.
6. Familiarity with AI, cloud computing, and cybersecurity.



NEEDS AND WANTS OF THE COMPUTER SCIENCE ENGINEER IN TODAY'S WORLD

- **WANTS (DESIRABLE GOALS)**

1. High-paying job in a reputed IT company.
2. Opportunities to work on advanced technologies like AI and robotics
3. Good work-life balance.
4. Career growth and promotions.
5. Opportunities to work abroad or remotely.
6. Professional certifications and higher studies.



ROLE OF COMPUTER SCIENCE ENGINEERING WITH OTHER ENGINEERING DISCIPLINES

ENGINEERING DISCIPLINES	ROLES OF COMPUTER SCIENCE ENGINEER	REAL WORLD PRODUCT
Civil Engineering	Smart infrastructure, traffic management	Smart City Systems
Mechanical Engineering	Automation, robotics, CAD/CAM software	Industrial Robots
Electrical Engineering	Smart grids, embedded systems	Smart Electric Meter
Electronics & Communication	IoT, embedded programming	Smart Home Devices
Automobile Engineering	Autonomous driving, vehicle software	Tesla Model 3

- **CASE STUDY**
SELF-DRIVING CAR

A self-driving car uses Artificial Intelligence (AI), sensors, cameras, and GPS to drive without human control. Computer Science Engineers develop the software that controls the vehicle and processes sensor data. They design AI and Machine Learning models to recognize roads, traffic signs, and obstacles. They also create navigation systems, ensure cybersecurity, and improve vehicle safety. Their work makes self-driving cars smarter, safer, and more reliable for future transportation.



MISCONCEPTIONS ABOUT COMPUTER SCIENCE ENGINEER

- 1.Computer Science Engineers only do coding. Reality: They also work in AI, cybersecurity, cloud computing, data science, networking, software testing, and system design.
- 2.They only repair computers. Reality: Hardware repair is usually done by technicians, while CSE engineers develop software and computing systems.
- 3.They must be experts in mathematics only. Reality: Logical thinking, problem-solving, and creativity are equally important.
- 4.They work alone all the time. Reality: Most projects require teamwork with engineers, designers, and clients.
- 5.A CSE degree guarantees a high-paying job. Reality: Success depends on skills, projects, internships, and continuous learning

SKILLS REQUIRED FOR COMPUTER SCIENCE ENGINEER IN 21ST CENTURY

1. TECHNICAL SKILLS

- Programming (C, C++, Java, Python)
- Data Structures and Algorithms
- Database Management (SQL)
- Artificial Intelligence (AI) and Machine Learning (ML)
- Cloud Computing
- Cybersecurity



2.SOFT SKILLS

- Problem-solving and Critical Thinking
- Communication Skills
- Teamwork and Collaboration
- Creativity and Innovation
- Time Management
- Leadership



- **AI'S USED FOR PRESENTATION**
 - **CHAT GPT (CONTENT)**
 - **CANVA (TEMPLATE)**
 - **PINTEREST (IMAGES)**

THANK YOU.....

BY.....

HARIPRASANTH S

// YEAR CSE